

THE BUILD

Whose HLA is in your reference?

Five projects. One graph of the MHC, a highly variable immune region.

754

MHC haplotypes

390 Asian or Arab haplotypes

752 biological haplotypes + 2 references

APR / HPRC r2 / JaSaPaGe / K-PanRef / CPC



Extract MHC → annotate → build graph (schematic)

A SHARED RESOURCE

Graph + allele labels

A WORKING PIPELINE

CWL workflow

A TEST BED

Reads: 2,504 samples

THE PAYOFF

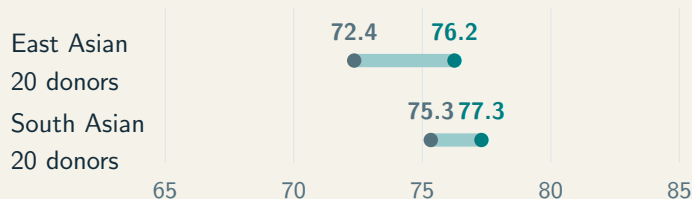
More haplotypes. Better SV calls.

Better recovery of genotypes carrying structural variants.

SV-bearing genotypes recovered exactly (%)

● HPRC-only panel

● Expanded panel



+3.9

points in East Asian donors

95% CI: +2.4 to +5.3

South Asian: +2.0 points

95% CI: +0.3 to +3.5

The clearest gain: recovering harder variants.

40-donor pilot, PanGenie, same reads; assembly-derived truth. Panel size and ancestry change together.

Test families excluded from panels; test assemblies remain in graph topology. All eligible SV-bearing genotypes.

THE SURPRISE

Better sequences. Same names.

A small change to SpecHLA: add pangenome sequences when collecting reads.

Exact gene reconstructions (out of 128)

8 donors $\times$ 8 genes $\times$ 2 haplotypes; assembly truth

Standard SpecHLA

41

+ pangenome in read binning

51

10 more exact sequences; same allele calls (61/63).

LOCITYPER / 40 DONORS

T1K leads on names.

Classical genes, 40 donors:

T1K **98.4%** / our panel **92.5%**

DRB3/4/5 type + copy number:

our panel **111/113 correct**
Stage B; before exon grafting.

Next: take this beyond the pilot.

Locked validation: 228 leave-one-out donors + 946 experimentally typed donors.